Rustambek Urokov

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Summary

Undergraduate researcher working on foundation models, representation learning, and protein language models, with interests in computational biology, speech technology, and AI safety. Python and PyTorch.

Manuscripts

- **Sequence-Derived Representations versus Pfam-Domain Content for Biosynthetic Gene Cluster Retrieval.** R. Urokov, A. Khan, F. Eshboyev, D. Asadov, S. Rahman, D. Kushokova. *First author. Preprint on bioRxiv (doi.org/10.64898/2026.08.21.746127), accepted as a poster at the ICBINB workshop, NeurIPS 2026; also submitted to Transactions on Machine Learning Research (TMLR), awaiting a decision.*

Research Projects

BGC-SetNet: Structure-Aware BGC Retrieval | *PyTorch, ESM-2, Set Transformers* **2026**

- Built a *Streptomyces griseus* atlas of 6,953 biosynthetic gene clusters and 316,257 proteins using NCBI, Prodigal, DeepBGC, BLASTp against MIBiG, and frozen ESM-2 embeddings.
- Designed BGC-SetNet, a Set Transformer trained with masked-gene reconstruction and family-level contrastive learning over protein embeddings and relative gene positions.
- Evaluated group-disjoint splits across five seeds and 16 held-out families, reporting Pfam Jaccard Recall@50 of 0.879 versus 0.847 for learned representations and documenting silver-label and power limitations.

ESM-2 Fine-Tuning for Protein Subcellular Localization | *PyTorch, HuggingFace PEFT/LoRA* **2026**

- Compared linear probing, full fine-tuning, and LoRA from 8M to 650M parameters for 10-class localization on 24,667 DeepLoc 2.0 proteins.
- Rank-16 LoRA on the 650M model reached 76.5% accuracy with 2.4% trainable parameters, matching full fine-tuning of the 150M model, with released checkpoints and a public demo.

Uzbek Speech Recognition | *PyTorch, HuggingFace Transformers, Whisper* **2026**

- Fine-tuned Whisper-medium on the 52-hour FeruzaSpeech corpus with 22,432 augmented samples, reaching 9.64% WER and 1.60% CER on the held-out test split.

Experience

Safia **2026 – Present**
Machine Learning Engineer *Tashkent, Uzbekistan*

- Working on applied machine learning systems at Safia.

Openbank **July 2026 – October 2026**
Machine Learning Engineer *Tashkent, Uzbekistan*

- Reviewed diffusion and flow-matching TTS papers, FLEURS and speech-dataset research, and TTS scaling studies to guide the OmniVoice model and data strategy.
- Built and shipped a self-hosted Uzbek OmniVoice text-to-speech system serving the Openbank mobile assistant.
- Fine-tuned OmniVoice on 220.75 hours and 148,529 clips for 50,000 steps on an RTX 5090, with leakage-free source-disjoint splits and an Uzbek text normalization front end.

Technical Skills

Languages: Python, SQL
ML: PyTorch, HuggingFace Transformers and PEFT/LoRA, scikit-learn
Speech & Audio: TTS fine-tuning, voice cloning, text normalization
Protein & Genomic Data: ESM-2 embeddings, FASTA, HDF5, Pfam annotations, MIBiG, DeepBGC, Prodigal, BLASTp
Tools & Infrastructure: Git, Linux, AWS, HuggingFace Hub and Spaces

Education

New Uzbekistan University **Oct. 2023 – June 2027**
Bachelor of Science in Economics and Data Science *Tashkent, Uzbekistan*

- Dean's List (top 10% of class), Spring 2026. IELTS 7.5 (C1) | SAT 1420.

Leadership & Achievements

IT Community of Uzbekistan **June 2023 – Present**

Project Management Volunteer *Tashkent, Uzbekistan*

- Coordinated logistics for technical events including Google I/O Extended for 100+ attendees, and interviewed Uzbek engineers at major tech companies for a community page reaching 5,000+ followers.

CAU Business Case Competition, Finalist **2025**

Mathematics for Machine Learning, DeepLearning.AI **2025**